

VARUN KANNEPALLI

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PROFESSIONAL EXPERIENCE

Geospatial Analyst , NASA , Greenbelt, MD

Jun 2022 — Aug 2022

- Developed algorithms for Geospatial data analysis, processing, and visualization.
- Created detailed reports and presentations for stakeholders.
- Utilized advanced GIS tools to enhance data accuracy.

EDUCATION

High School Diploma

May 2024

Wootton High School, Rockville, MD

B.S in Mechanical Engineering

May 2028

University of Maryland, College Park

Relevant Coursework:

Finished : Introduction to Engineering Design (ENES100) Mechanics I (ENES102) Differential Equations (MATH246)
Dynamics (ENES221) Mechanics II (ENES220) Introduction to CAD (ENME272) Intro to Computing (ENME202)
Current: Thermodynamics (ENES232) Intro to Materials Engineering (ENME382)

TECHNICAL SKILLS

CAD (Solidworks, Autodesk)

Coding (Python, Java, C++)

Circuit Design and Soldering

PROJECTS

Over-Terrain Vehicle Project

Designed an Over-Terrain Vehicle (OTV) to navigate past obstacles and identify various substrate types (non-plantable vs plantable) using color sensors. OTV was fabricated by 3D printing/laser cutting and used Arduino programming, custom sensors, and motors.

Truss Design Project

Constructed a wooden truss to support over 1,000 lbs while adhering to strict constraints (size and weight). Conducted structural analysis, assessed internal forces, buckling, and safety factors. Used pine wood and Gorilla Glue for the truss, which achieved a predicted failure load of ~1318 lbs and a strength-to-weight ratio of 4903.

Beam Design Project

Designed a wooden I-beam to support over 1000 lbs while adhering to strict constraints (size and weight). Optimized Beam dimensions using a spreadsheet and MATLAB, developed a CAD model, and assessed the maximum possible failure mode. Using pine wood and Gorilla Glue for the beam, we predicted bending failure of 2,300 lbs and a strength-to-weight ratio of 4,440.

Phone Stand Cad Model/3D print

Made a phone stand in SolidWorks that could withstand the weight of my phone. It also had to weigh less than 20 grams and meet specific length and height requirements. Conducted static FE analysis calculating deflection, 3d printed the model in PLA material, and now regularly use the phone stand every day.

Solar Powered Portable Charger

Built a solar-powered portable charger as part of a team using a Li-Po battery, solar panel, rectifying diode, and USB booster module. Soldered the components to a circuit board and applied heat shrink tubing for safe and efficient charging.

OTHER ACTIVITIES

First Lego League (Competitor and Mentor)

2017 — 2023

- Competed for four FLL seasons, earning the Champions Award, the Best Robot Design Award in 2018, and the Global Innovation Award in 2019.
- Mentored FLL team "CCC Builders" on robot design and strategy, leading to a first-place finish at the 2023 Bowie Qualifier competition.

Carnatic Music Vocalist

Present

- Performed Carnatic Music Concerts around the DC Metro Area and India with my twin brother.
- Recently, we performed 7 concerts in India as part of the Margazhi Festival